PHILOMATH NETWORKS

PHILOMATH NETWORKS PVT. LTD

6-Days Hands-On Workshop on

" Data Science Certification Training Course"

Get ready to earn your Data Science Certification with us!

Redefining Learning
Beyond Boundaries

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About Philomath Networks

At Philomath Networks Pvt. Ltd., we specialize in organizing transformative events for schools, organizations, and communities, fostering a culture of exploration, innovation, and growth. From expert-led workshops to competitive events, we create immersive learning experiences that inspire and empower individuals to go beyond boundaries. Partner with us to bring impactful learning to your community!

Partner With Us

Collaborate with Philomath Networks to bring innovation, engagement, and excellence to your institution or organization. Let's work together to create an ecosystem of lifelong learning and impactful experiences!

What Sets Us Apart?

Comprehensive Learning Approach – We blend academic, non-academic, and skill-based experiences through workshops, competitions, expert-led sessions, and immersive programs tailored to real-world applications.
End-to-End Event Management – From conceptualization to execution, we ensure seamless organization of competitions, tech fairs, hackathons, leadership programs, and customized training sessions that align with institutional goals.
Industry & Academic Bridging – We foster collaborations between institutions, corporates, and experts, ensuring that participants gain practical exposure, networking opportunities, and insights into emerging trends beyond traditional education.

Customizable & Scalable Programs – Whether it's a school-level science fest, university hackathon, corporate leadership training, or specialized certification courses, we tailor our events to meet the unique needs of every institution.

Proven Impact & Engagement – Our interactive approach ensures higher participation, deeper learning retention, and enhanced skill development, making every event a transformative experience.

Course
Overview

Course Objective

This intensive training program provides a comprehensive understanding of Data Science concepts, tools, and techniques. Participants will explore data preprocessing, exploratory analysis, machine learning, deep learning, and real-world applications using Python, SQL, and cloud-based solutions. Through hands-on projects and real-world case studies, learners will gain practical experience in data analysis, model building, and optimization. By the end of the course, participants will be proficient in handling and analyzing data, developing predictive models, and deploying solutions, preparing them for roles in data science, machine learning, and analytics.

Course Outcome

- Data Preprocessing & Analysis – Clean, transform, and visualize
- NLP & Time Series – Analyze text and forecast trends.
- Machine Learning Models Build and evaluate
- Model Optimization & Deployment
- Deep Learning Applications Implement neural networks
- Real-World Problem Solving – Work on industry projects.

Training
Modules

The training modules provide a hands-on learning experience in core Data Science concepts, data analysis, and machine learning techniques. Participants will begin with data fundamentals, preprocessing, and exploratory analysis using Python, SQL, and visualization tools. They will then explore machine learning, deep learning, NLP, and time series forecasting to build predictive models and extract insights. Each module is structured around real-world applications, ensuring practical skill development. Additionally, learners will gain expertise in optimizing models, managing large datasets, and deploying data-driven solutions for enhanced analytics efficiency.

Module 1: Foundations of Data Science

- Introduction to Data Science – Concepts, lifecycle, and industry applications
- Python for Data Science – Basics, libraries (NumPy, Pandas, Matplotlib)
- Data Collection & Storage – Working with CSV, JSON, databases
- SQL for Data Science – Querying and managing structured data
- Data Cleaning & Preprocessing – Handling missing values, outliers, and inconsistencies
- Exploratory Data Analysis (EDA) – Visualizing trends and patterns

Module 2: Machine Learning Fundamentals

- Supervised vs Unsupervised Learning – Key differences and applications
- Regression Models – Linear and logistic regression basics
- Classification Techniques – Decision trees, SVM, KNN
- Clustering Algorithms – K-Means, hierarchical clustering
- Model Evaluation – Accuracy, precision, recall, F1-score
- Hyperparameter Tuning – Grid search, cross-validation

Module 3: Advanced Machine Learning & Deep Learning

- Ensemble Methods – Random Forest, Gradient Boosting, XGBoost
- Neural Networks – Introduction to perceptrons and activation functions
- Deep Learning Architectures – CNN, RNN, and LSTMs
- Transfer Learning – Leveraging pre-trained models
- Optimizing Model Performance – Overfitting, regularization, dropout
- Deploying ML Models – Flask, FastAPI, cloud deployment

Training Modules

As the course progresses, participants will explore data pipeline automation, model deployment, and cloud-based analytics. They will learn monitoring techniques using MLflow and Prometheus and understand security best practices in data science workflows. The final modules focus on data governance, model optimization, and ethical AI, ensuring learners are industry-ready to implement scalable, secure, and efficient data-driven solutions. By the end of the training, participants will have the confidence to design, deploy, and manage machine learning models in real-world environments, preparing them for data science roles in leading tech industries.

Module 4: Natural Language Processing (NLP) & Time Series Analysis

- Text Preprocessing – Tokenization, stopword removal, stemming, lemmatization
- Feature Extraction – TF-IDF, Word2Vec, BERT embeddings
- Sentiment Analysis – Building and evaluating sentiment classifiers
- Topic Modeling – LDA, NMF for text clustering
- Time Series Forecasting – ARIMA, LSTMs, Prophet
- Real-World Applications – Chatbots, stock market predictions

Module 5: Data Engineering & Big Data Analytics

- Big Data Overview – Hadoop, Spark, NoSQL databases
- Building Data Pipelines – Apache NiFi, ETL processes
- Cloud Computing in Data Science – AWS, Google Cloud, Azure
- Monitoring & Security – MLflow, Prometheus, data privacy best practices
- Data Governance & Compliance – GDPR, ethical AI considerations
- Optimizing Workflows – Cost reduction, scalability strategies

Module 6: Real-World Applications & Capstone Project

- Industry Case Studies – Fraud detection, recommendation systems
- End-to-End Model Deployment – Integrating data science solutions
- AI in Business Decision-Making – Forecasting and data-driven insights
- Automation & MLOps – CI/CD for ML models
- Final Capstone Project – Solve a real-world data science problem
- Career Preparation – Resume building, interview tips, portfolio projects

Training Schedule & Format

The training schedule is structured as a 6-day intensive Data Science workshop, providing a hands-on learning experience. Each day focuses on a key Data Science module, starting with foundational concepts and progressing to advanced topics like machine learning, deep learning, NLP, and big data analytics. The workshop includes interactive sessions, hands-on labs, and real-world problem-solving to reinforce learning. Participants will work on mini-projects and practical assignments daily, ensuring a strong grasp of data science tools and techniques. The final day is dedicated to a capstone project, model optimization, and a certification assessment, equipping learners with industry-ready Data Science skills.

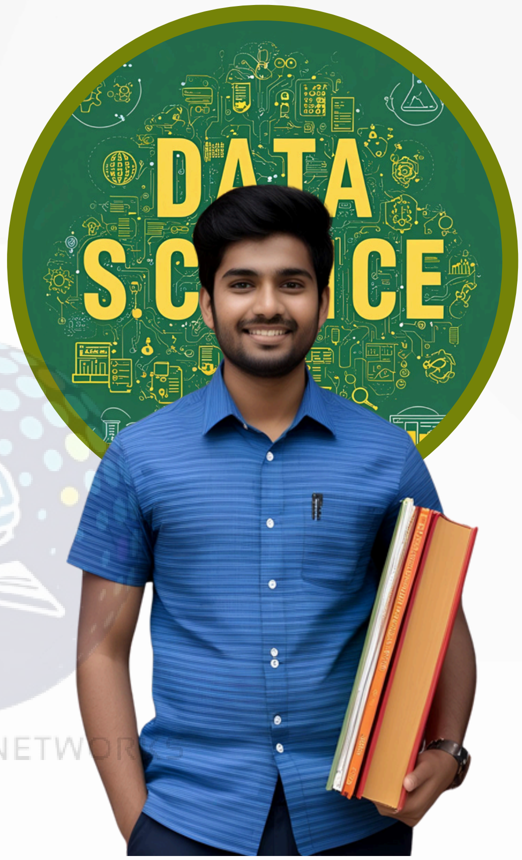
Day	Module	Focus Area	Duration
Day 1	Foundations of Data Science	Data Handling	5hrs
Day 2	Machine Learning Fundamentals	Model Building	5hrs
Day 3	Advanced ML & Deep Learning	Neural Networks	5hrs
Day 4	NLP & Time Series Analysis	Text & Forecasting	5hrs
Day 5	Data Engineering & Big Data Analytics	ETL & Security	5hrs
Day 6	Capstone Project & Certification	Project & Exam	5hrs

Through hands-on exercises and case studies, participants will develop essential Data Science skills, explore best practices, and tackle real-world data challenges. The workshop includes live demonstrations, guided projects, and interactive labs to enhance learning. A collaborative format encourages discussions, troubleshooting, and peer learning. By the end, attendees will have a strong Data Science foundation, enabling them to analyze large datasets, build optimized models, and deploy scalable data solutions effectively.

How to Get
Involved

Steps to Enroll in Training Programs

- Contact us via email or phone.
- Share your preferred course, schedule, and any specific requirements.
- Our team will review your request and confirm the booking.
- You will receive confirmation details via email or call .



Contact
Information

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